VICTORY 2027

Soil Resources

LECTURE-2 Geography

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Topics To Be Covered

1. Types of soils
2. Laterite Soil
3.
4.

Soil Resources

- *Types of soil (alluvial, black, red and laterite), mode of formation, distribution, composition and characteristics such as colour, texture, minerals and crops associated.*
- *Soil Erosion – meaning, agents and causes.*
- *Soil Conservation – importance and methods*



Captain India!

Telegram

https://t.me/icsebysameersir



Ex-situ

Q13. [PYQ] Inland Alluvial soils of the Indo-Gangetic plains are 'ex-situ' soils. What does 'ex-situ' mean?

- A) Formed in their original place of weathering
- B) Transported by agents like rivers and winds from their place of origin
- C) Lacking any mineral content
- D) Formed due to volcanic action

QUESTIONS

Organic matter derived from the decomposition of dead plants and animals by micro-organisms, which provides nutrients and moisture-retention capacity to the soil, is called:

Humus

1. [illegible]
2. [illegible]
3. [illegible]
4. [illegible]

4. The original rock material from which a particular soil is derived, determining its mineral composition, color, and texture, is called the:

A) Topsoil

B) Subsoil

C) Parent Rock

D) Regolith

QUESTIONS

Which soil is also known as Riverine Soil

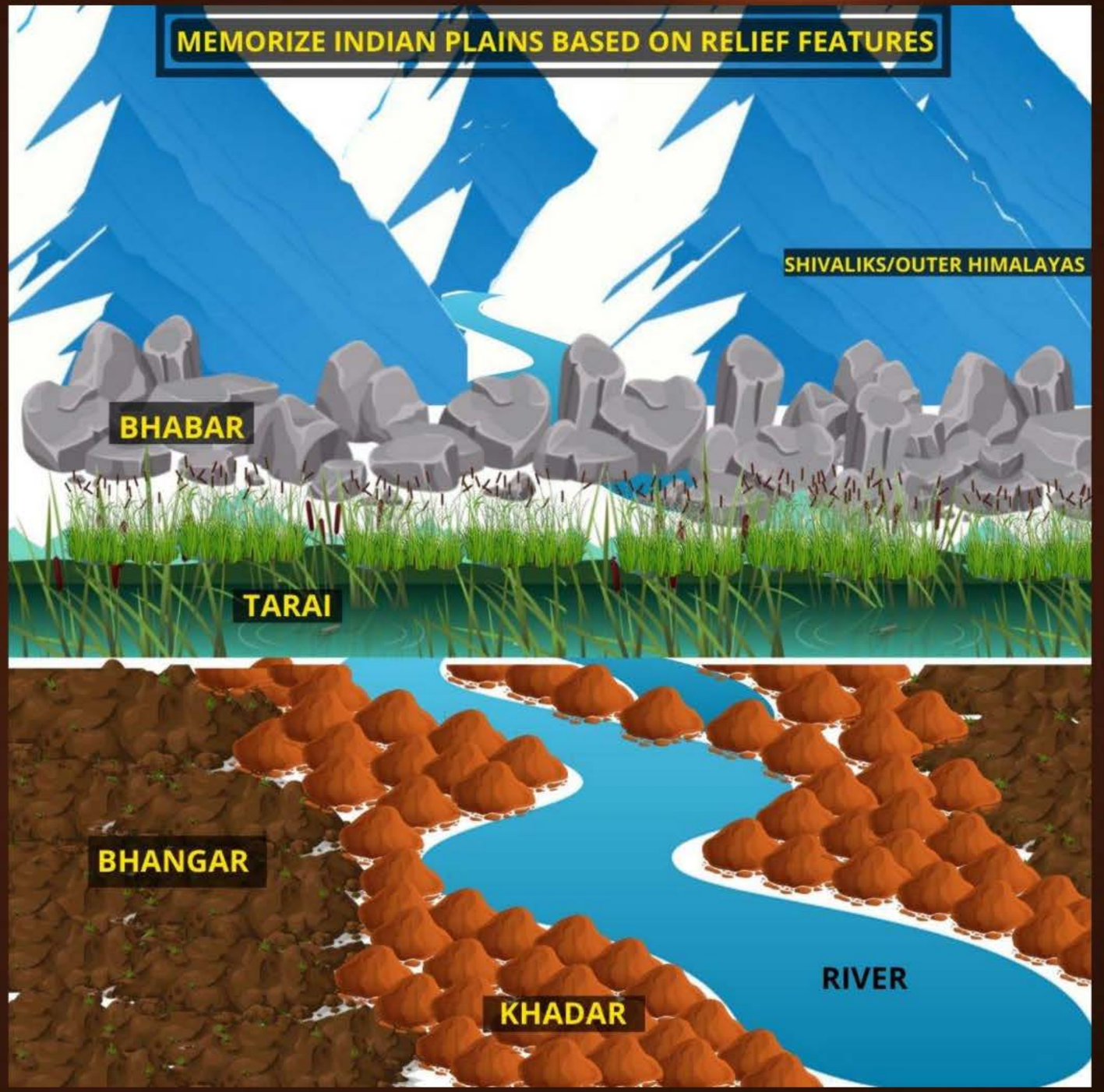
1. Black Soil
2. Red Soil
3. Alluvial Soil
4. Laterite Soil

Q3. [PYQ] Differentiate between Khadar and Bhangar. Khadar soil is:

- A) Old, coarse, and found in upland areas
- B) New, fine, fertile, and frequently renewed by floods
- C) Clayey, dark, and moisture-retentive
- D) Leached, acidic, and rocky


Bhangar
Khadar




MEMORIZE INDIAN PLAINS BASED ON RELIEF FEATURES
SHIVALIKS/OUTER HIMALAYAS
BHABAR
TARAI
BHANGAR
RIVER
KHADAR

BLACK SOIL




INDIA
MAJOR SOIL TYPES
N
68°E
72°
76°
80°
84°
88°
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36°N
32°
28°
24°
20°
16°
12°
8°N
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
Tropic of Cancer
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
SRI LANKA
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
Forest & Mountainous
Alluvial
Red and Yellow
Black
Laterite
Arid
0 200 400 600 km



2. Black Soil

- This soil is black in colour and is also known as the Black Cotton Soil or Regur soil.
- It is dark in colour and is suitable for cotton cultivation.
- This soil is the residual soil, i.e., it is formed at the place of its origin over the underlying rocks.
- Since it is formed by the denudation of volcanic rocks, it is also known as lava soil.

→ Breaking down

2. Black Soil

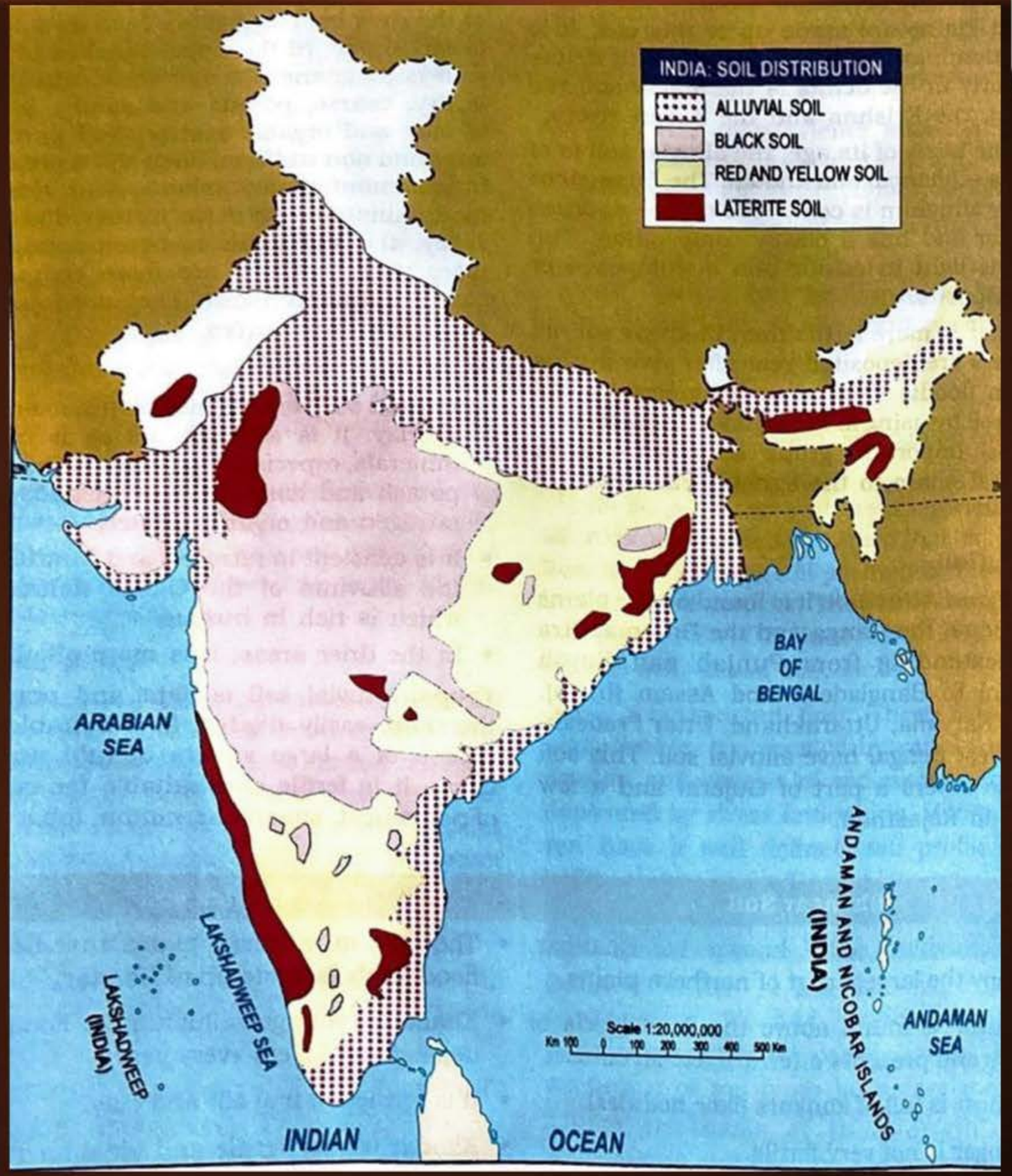
Distribution

- The Regur soils are concentrated over the Deccan lava tract which include parts of Maharashtra, Madhya Pradesh, Gujarat, Andhra Pradesh, Karnataka and some parts of Tamil Nadu.

2. Black Soil

Characteristics of Regur (Black) Soil

- **Colour:** It is black in colour as it is formed from weathered lava rocks.
- **Texture:** It is fine textured and clayey in nature (up to 50% clay content) and, therefore, highly retentive of water.
- Because of high clay content, this soil expands when wet and becomes difficult to plough.
- During the dry season, it shrinks and develops big cracks which help in air circulation. This characteristic causes a kind of 'self-ploughing'.
- Due to slow absorption and loss of moisture, the black soil retains moisture for a very long time which helps the crops, especially, the rain fed ones to sustain even during the dry season.


INDIA: SOIL DISTRIBUTION
ALLUVIAL SOIL
BLACK SOIL
RED AND YELLOW SOIL
LATERITE SOIL
ARABIAN SEA
BAY OF BENGAL
LAKSHADWEEP SEA
LAKSHADWEEP (INDIA)
ANDAMAN AND NICOBAR ISLANDS (INDIA)
ANDAMAN SEA
INDIAN OCEAN
Scale 1:20,000,000
Km 100 0 100 200 300 400 500 Km

2. Black Soil

- **Minerals:** It has high quantities of lime, iron, magnesium and generally poor percentage of phosphorous, nitrogen and organic matter.
- **Crops:** The soil is very fertile in most of the places.
- It is suitable for cultivation of cotton, jowar, wheat, linseed, gram, citrus fruits and vegetables, tobacco and sugarcane.
- In any season it has moisture stored in its subsoil.
- The moisture retentiveness of black soil makes it suitable for dry farming.

2. Black Soil

Dry farming refers to crop cultivation without irrigation during the dry season as the crops would utilize the moisture stored in the soil during the rains.


Red SOIL

3. Red Soil



- Red soil is a category of soil which develops on old crystalline rocks.
- Under prolonged weathering by rainfall, ancient crystalline and metamorphic rocks of the Peninsular plateau break up to form this soil.
- This soil differs from place to place on the basis of the parental rock material and climatic conditions.


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3. Red Soil

Distribution

- Red soil forms the second largest soil group in India and is found mainly on the plateau region of Peninsular India from Tamil Nadu in the south to Bundelkhand in the north and Rajmahal Hills in the east to Kutch in the west.

- Red soil covers almost the whole of Tamil Nadu, Karnataka, Andhra Pradesh, south-eastern Maharashtra, parts of Odisha, Chhattisgarh, Jharkhand, Bundelkhand, Meghalaya, Mizoram, Manipur, Telangana and Nagaland.

3. Red Soil

Red → Crystalline
↓
Rust (जंग) { Iron + Oxygen
↓
Reddish

- Scattered patches of red soil are also found in Birbhum (West Bengal), Mirzapur, Jhansi, Banda, Hamirpur (Uttar Pradesh), Udaipur, Chittorgarh, Dungarpur, Banswara and Bhilwara districts (Rajasthan).
- It practically encircles the entire black soil region on all sides.
- It extends northwards in the west along the Konkan Coast of Maharashtra.

3. Red Soil

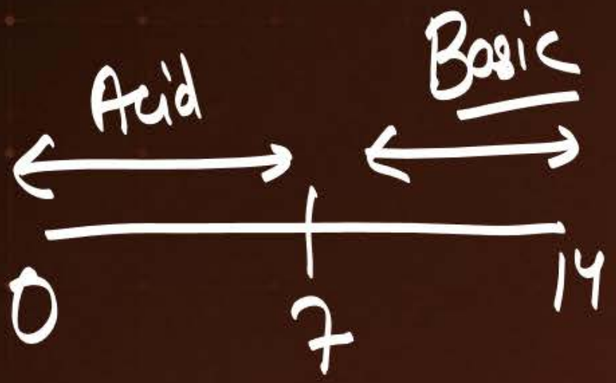
Crystalline rocks
↓
Iron + oxygen
↓
Red. Brown

Characteristics of Red Soil

- **Colour:** It is red in colour as it contains large amounts of iron oxide.
- At several places, its colour slightly changes and it appears brown or grey.
- It looks yellow when it occurs in a hydrated form.

Iron + oxygen + water
↳ yellow

3. Red Soil



Air can easily pass by

- **Texture:** It is porous, loose and aerated. It is generally shallow. Its pH value ranges from 6.6 to 8.0.
- **Minerals:** It is poor in nitrogen, phosphorus, potassium and organic matter. It contains soluble salts in small quantities.

3. Red Soil

- **Crops:** Its porous nature makes red soil ideal for dry farming.
- The limited rainfall can percolate and be stored in the subsoil from where it can be utilised by the roots of plants.
- It needs irrigation and fertilizer support for crop cultivation.
- It is suitable for cultivating almost all crop types, it is most suitable for growing vegetables, rice, ragi and tobacco.
- Groundnut and potatoes can be grown on coarse soils at higher levels and sugarcane on heavy clays at lower levels.

5. Black soil is formed due to the weathering of which type of rocks?

A) Metamorphic rocks → Crystalline → Red Soil

B) Basaltic lava rocks

C) Sedimentary sandstone

D) Granitic gneisses

8. Red soil develops its distinctive reddish color due to the presence of:

A) High organic content

B) Hydrated iron oxides

C) Wide diffusion of iron in crystalline and metamorphic rocks

D) Excessive leaching of silica

Day-Cultivation

Q5. [PYQ] Black soil is known for its 'self-ploughing' characteristic because:

- A) It is rich in organic matter.
- B) It develops deep cracks when dry, allowing oxygenation.
- C) It does not require water for cultivation.
- D) It is easily eroded by wind.

QUESTIONS

Which soil is also known as Regur Soil

1. Black Soil
2. Red Soil
3. Alluvial Soil
4. Laterite Soil

Alluvial Soil
- Riverine Soil
- Deposition of rivers.
  - Bhangar
  - Khadar

Black Soil
- Lava Rocks, Volcanic, Igneous Rocks.
- Regur, Cotton Soil
- Fine-clayey.
- Self ploughing
- Dry Farming.

Red Soil

- Crystalline or metamorphic rocks.
- Iron Content + Oxygen → Reddish
- Iron + Oxygen + water → Yellowish


Laterite Soil

4. Laterite Soil

High rainfall
High temperature

Disilication
Leaching
Eluviation
} Washing away of nutrients of soil due to heavy rainfall.

- This soil type is formed as a result of atmospheric weathering of rocks under conditions of high rainfall and temperature with alternate wet and dry periods.
- It is the residual soil formed by leaching due to tropical rains.
- Leaching is the process in which the nutrients get percolated down below the soil due to heavy rainfall; thus leaving the topsoil infertile. This is also called desilication.
- Due to heavy rains, lime and silica are leached away and aluminium compounds are left behind.
- Humus content of the soil is removed by bacteria that survives well in high temperature.

4. Laterite Soil



- There are two types of laterite soils: Upland Laterites and Lowland Laterites.
- Upland laterites are formed over hills and uplands.
- From there they are transported by streams towards lowlands.
- Such transported soils are known as Lowland Laterites.

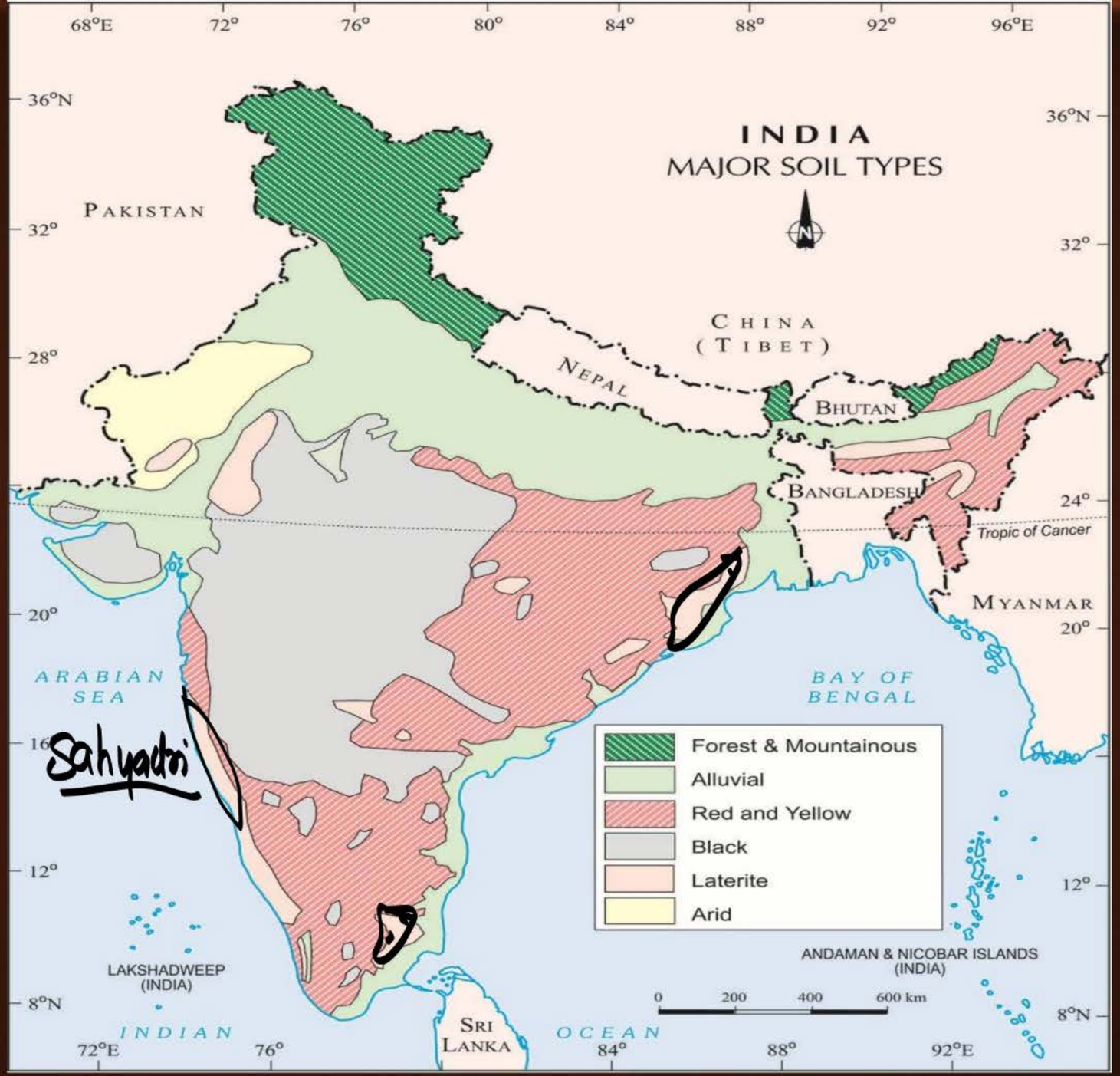
4. Laterite Soil

Summit → Peak

Distribution

- Laterite soil mainly occurs in the highland areas of the Peninsular Plateau, especially on the summits of the Sahyadris, Eastern Ghats, Rajmahal Hills and many other hills in the eastern parts of the peninsula.

- Patches of laterite soil are found in Madhya Pradesh, Odisha, Maharashtra, West Bengal, Andhra Pradesh, Telangana, Karnataka, Kerala, Tamil Nadu and along the top slopes of the peninsular mountains, the Western and Eastern Ghats.


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4. Laterite Soil

Characteristics of Laterite Soil

- **Colour:** It is red in colour due to the presence of iron oxide which is formed by leaching.
- The soluble plant foods like potash are removed from the topsoil leaving alumina and iron oxide.
- **Texture:** It is of a coarse texture, soft and breakable

↳ Rough

4. Laterite Soil

- **Minerals:** It is a porous soil; silica is removed from it by chemical action.
- It is poor in lime and magnesium and deficient in nitrogen.
- It is acidic in nature as alkalis (Base) are leached. It is rich in iron oxide.
- It is widely cut as bricks and is considered suitable for construction materials.

4. Laterite Soil



- **Crops:** Being infertile it suits only special crops like, cashew nuts, tea, coffee, etc.
- It has low fertility because of high acidity and low moisture retention, manuring and other activities are required to make it suitable for growing crops such as ragi, rice and sugarcane.
- Paddy is grown on lower elevations, whereas tea, cinchona, rubber and coffee are grown on higher elevations.

Soil	Formation	Areas	Characteristics	Crops
Alluvial Soil	Deposition of sediments by rivers.	Inland alluvium in Punjab, Haryana, U.P., Bihar, West Bengal, parts of Gujarat and Rajasthan. Deltaic alluvium in the deltas of Ganga-Brahmaputra, Mahanadi, Godavari, Krishna and Kaveri. Coastal alluvium along the coastal strips of the Peninsula.	Loamy. Coarse and dry in upper reaches of the river and gets finer and moist as the river flows. Rich in minerals especially potash and lime. Poor in nitrogen and humus.	Large variety of Rabi and Kharif crops; rice, wheat, sugarcane, cotton, gram and oilseeds; jute in GangaBrahmaputra delta.
Black Soil	Residual soil formed by weathering of lava rocks.	Deccan lava tract. Maharashtra, Madhya Pradesh, Gujarat, Andhra Pradesh, Karnataka and parts of Tamil Nadu.	Clayey. Black in colour. Rich in lime, Magnesium. Poor in phosphorous, nitrogen and organic matter. Very fertile.	Cotton, cereals, citrus fruits and vegetables, tobacco, and sugarcane.
Red Soil	Prolonged weathering of crystalline rocks. Differs on the basis of parent rock material and climatic conditions.	Plateau region of Peninsular India extending northwards along Konkan coast. Tamil Nadu, Karnataka, Andhra Pradesh, South-East Maharashtra, Chhattisgarh, parts of Odisha, Jharkhand, Bundelkhand, Meghalaya, Mizoram, Manipur, Telangana and Nagaland.	Ranges from Sandy to Clayey. Red in colour due to large amounts of ironoxides Deep and fertile in lowlands; thin and poor in highlands. Poor in nitrogen, phosphorus and organic matter.	Vegetables, rice, ragi, tobacco, groundnut and potatoes.
Laterite Soil	Due to leaching in areas of heavy rain.	Highland areas of Peninsular plateau. Patches in Madhya Pradesh, Odisha, Maharashtra, West Bengal, Andhra Pradesh, Telangana, Karnataka, Kerala, and Tamil Nadu.	Coarse and porous. Red due to Iron Oxide. Poor in lime, nitrogen and magnesium. High acidity and low moisture retention.	Tapioca, cashew nuts. With manure ragi, rice, sugarcane, tea, rubber and coffee.

Q9. [PYQ] Laterite soil is formed under conditions of:

- A) Low temperature and low rainfall
- B) High temperature and heavy rainfall with alternate wet and dry periods
- C) Low temperature and high humidity
- D) High temperature and dry desert climate

Q10. The process by which nutrients are washed away from the top layer of Laterite soil due to heavy rainfall is called:

- A) Humification
- B) Weathering
- C) Leaching (or Eluviation)
- D) Capillary action


Saturated

Thank You